

Certified Artificial Intelligence Professional (CAIP) Final Project Guidelines

Purpose:

The Final Project is a capstone requirement for the Certified Artificial Intelligence Professional (CAIP) program. It allows participants to demonstrate their ability to apply AI concepts, tools, and techniques to a practical problem aligned with industry and societal needs.

1. General Requirements

- a. **Project Type:** Applied AI project (data-driven, model development, or AI-enabled system).
- b. **Scope:** Must cover at least **two CAIP modules** (e.g., Machine Learning + IoT, Deep Learning + NLP, Generative AI + Governance).
- c. **Dataset:** Use a **real dataset** (open-source, industry-provided, or self-collected).
- d. **Tools:** Python (preferred), with libraries such as TensorFlow, Keras, scikit-learn, Pandas, Hugging Face, etc.
- e. **Report Length:** **5–8 pages** (excluding references, figures, and appendices).
- f. **Submission Format:** PDF

2. Structure of the Report

- a. **Title Page:** Project Title, Group Member Names, Submission Date
- b. **Abstract (150-200 words):** Summary of objectives, methods, dataset, and results
- c. **Introduction (1-2 pages):** Problem background, motivation, and objectives
- d. **Literature Review (1-2 pages):** Optional
- e. **Methodology (2-3 pages):** Dataset description, preprocessing, models used, workflow diagrams
- f. **Implementation & Results (3-4 Pages):** Experiments, performance metrics, graphs, and key findings
- g. **Discussion & Conclusion (1-2 Pages):** Interpretation of results, challenges, limitations
- h. **References:** 5-10 scholarly/industry sources in IEEE format.

3. Evaluation Criteria

Component	Weight (%)
Problem Definition & Relevance	10
Methodology & Workflow Design	25
Implementation & Technical Accuracy	30
Results & Analysis	25
Report Quality (writing, formatting, references)	10

4. Topics

Participants may select from the following general domains or propose a topic of their own. While suggested topics are provided for guidance, participants are strongly encouraged to select a project that aligns with their personal interests, professional background, or an area they wish to explore further. This flexibility is intended to allow participants to gain maximum value by working on a problem that is both meaningful and relevant to their future career or research goals.

- Predictive Maintenance in Manufacturing
- AI for Demand Forecasting & Inventory Optimization
- Generative AI for Automotive/Industrial Design
- AI in Healthcare (Image or Text Data)
- AI in Governance & Policy Analysis

5. Rules for Project Proposal

- Must have clear problem statement and real-world dataset. (Must cover at least two CAIP modules)
- Must evaluate performance using appropriate metrics.
- Must connect findings to real-world industry or governance relevance.

6. Deliverables & Timeline

- Project Proposal:** A one-page summary, prepared in accordance with the rules outlined in **Section 5**. The proposal must be submitted no later than **July 28, 23:59 hrs**.
- Final Report:** A comprehensive report, developed as per the instructions in **Section 2**, with the complete code file attached as an annex. The Python source file shall also be

submitted separately for testing and verification. The final report and accompanying materials must be submitted no later than **August 25, 23:59 hrs.**

7. Group Formation

You can work on the project individually or in a group of up to **two members maximum.**

8. Submission Instructions

For Proposal Submission:

Email: lab.tech@ncai.nust.edu.pk

CC: rmeo@ncai.nust.edu.pk

Subject: CAIP Proposal Batch 8

For Final Project Submission:

Email: lab.tech@ncai.nust.edu.pk

CC: rmeo@ncai.nust.edu.pk

Subject: CAIP Final Project Batch 8

